

Predictability and decomposability separately contribute to compositional processing of idiomatic language

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Abstract

When reading, comprehenders construct a message-level representation and integrate new information as it becomes available. Such compositional processing may differ for idioms, where the meanings of the individual words do not always relate to the figurative meaning. Here, we examined how predictability and idiom decomposability contribute to compositional processing. Participants' EEG was recorded while they read sentences containing idioms that varied in decomposability and phrase-final word cloze probability, or their literal match (break the ice/slip on the ice) with little context prior to the phrase, along with adjective insertion conditions (break the freezing ice/slip on the freezing ice). Cloze probability modulated N400 amplitudes to critical words for both idiomatic and literal phrases, whereas P600 amplitudes only elicited by idiomatic phrases were also modulated. Phrases with adjective insertions reduced acceptability judgments, particularly for idioms, and led to N400 amplitude differences compared to critical words, but only for idioms. N400 differences were also found between idiomatic and literal contexts at the point of adjective insertion. Additionally, both cloze probability and decomposability modulated gamma band activity, with greater gamma activity for more predictable and less decomposable idioms, but at different times and with different scalp topographies, suggesting dissociable components of processing. These results support a hybrid model of processing in which multiple linguistic factors determine the type of processing engaged by the brain to comprehend non-literal language. When prior context is minimal, compositional processing may still be engaged when reading idioms.

KEYWORDS

ERPs, idiom, psycholinguistics, sentence processing, time-frequency analysis

1 | INTRODUCTION

As a person reads, the meanings of individual words accrue incrementally to construct a message-level meaning

(Van Berkum, 2008). In literal language, the meanings of the lexical and phrasal constituents add up to the meaning of the whole. However, in the case of idioms, form-to-meaning mapping is often opaque, and some degree of prior

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experience or knowledge may be needed to understand the figurative meaning. Several factors may influence how idioms are processed; for instance, predictability of words in the idiom (Titone & Connine, 1994) may play a role, as it has been well established that predictability facilitates sentence processing (Federmeier, 2007). Decomposability, or the degree to which an idiom's meaning can be derived from the semantics of its component parts, can also influence idiom comprehension (Coulson, 2008). For example, some idioms are opaque, in that the literal meanings of the constituent words have little to do with the figurative meaning (e.g., *be on cloud nine*), and thus have low decomposability. Others are more transparent, and the figurative meaning can be derived from the meanings of the individual lexical parts (e.g., *be a slap in the face*), resulting in higher decomposability. There are currently mixed findings on whether increased decomposability facilitates or inhibits non-literal meaning comprehension (Cacciari & Tabossi, 1988; Gibbs, 1980; Nunberg et al., 1994; Swinney & Cutler, 1979). Additionally, very few studies have investigated neural processing of idioms taking decomposability into account (Cieślicka, 2013). Here, we employed electrophysiological recordings to investigate how processing differs between idioms and literal expressions during online comprehension, and how decomposability and predictability modulate these differences.

1.1 | Theories of idiom processing

Theories of idiom processing differ on whether idioms are processed holistically or compositionally at first during comprehension. *Noncompositional* models suggest that idioms' meanings are stored and retrieved from semantic memory as chunks, meaning that constituent words that make up the idiom are not fully processed (Bobrow & Bell, 1973; Gibbs & Gonzales, 1985). Such holistic storage and retrieval of idiomatic meaning would entail an advantage in processing speed for idioms compared to literal phrases, as the meaning can be activated in its entirety all at once, lessening the need to process the constituent words. Swinney and Cutler (1979) demonstrated this idiom advantage in a classification task, where participants were faster to indicate that idiom items were meaningful phrases compared to literal controls. Similarly, Gibbs (1980) showed that when idioms were intended with their expected, nonliteral sense, processing was facilitated compared to when intended with a literal interpretation. However, memory for literal usage of idioms was greater than for conventional, idiomatic usage, as participants required further elaboration of the literal usage due to their unfamiliarity, leading to better memory. According to noncompositional theories, lexical

properties of words may have less impact when embedded in an idiomatic phrase; namely, once sufficient information has been obtained to identify the idiom, the idiomatic meaning would be retrieved and the initial literal processing of the constituent words would cease. Indeed, some experimental work has demonstrated that word concreteness (Peterson et al., 2001) and semantic congruity (Rommers et al., 2013) are not processed when in idiomatic phrases compared to literal phrases. This theoretical framework potentially suggests that predictability of words of an idiom or decomposability of the idiomatic phrase would have little impact on the processing speed of the idiom, since the constituent words do not undergo full processing.

Conversely, *compositional* models posit that each of an idiom's component lexical parts are initially processed compositionally, with meaning accruing over time as a phrase is read, similar to literal language processing (Gibbs et al., 1989; Sprenger et al., 2006; Titone & Connine, 1994). These models suggest that idiomatic meaning activation requires recognition of a configuration of specific lexical items in a particular order. The *configuration hypothesis* suggests that within every idiom is a *recognition point* at which a person recognizes they are reading an idiom (Cacciari & Tabossi, 1988). Once this recognition point is reached, nonliteral meaning is activated directly from memory. Titone and Connine (1994) argued in support of the configuration model with a series of cross-modal priming experiments showing that item properties such as predictability modulated idiom meaning activation. Furthermore, if an idiom was literally plausible (e.g., *be a slap in the face*), the idiom's literal meaning was still primed, suggesting that the idiom component parts undergo full semantic analysis. Other language production work has demonstrated that semantic and syntactic production errors of idiomatic phrases did not differ from those of literal phrases, suggesting similar processing and representation of these types of information (Cutting & Bock, 1997). These models of idiom processing contrast with non-compositional models, and suggest that lexical properties of words would affect processing regardless of idiomatic or literal context, as literal analysis would occur in both cases.

Finally, other models take a hybrid approach, in which non-compositional versus compositional processing of idioms depends on the characteristics of the idiom itself (Cacciari & Tabossi, 1988; Libben & Titone, 2008; Titone & Connine, 1999). For instance, idioms that are more familiar or predictable may undergo less compositional processing, as they may be recognized or predicted, and then retrieved as a whole from memory. In contrast, unfamiliar or less predictable idioms may undergo full compositional analysis with nonliteral meanings only activating through

contextual processing. Additionally, the decomposability of an idiom may play a role in its comprehension. For instance, in highly decomposable idioms, the meanings of the constituent words relate to the meaning of the figurative phrase; thus, compositional analysis of decomposable idioms may be effective for figurative understanding. On the other end of the scale, the more non-decomposable an expression is, the less transparent the mapping between the constituent words and the figurative meaning, and thus more holistic and non-compositional processing may be a more effective processing strategy for accessing the figurative meaning. This difference may influence language processing; however, the few studies that have investigated this specific factor have produced mixed results. For instance, one study demonstrated that more decomposable idioms were processed faster than less decomposable idioms, suggesting a difference in processing (Caillies & Butcher, 2007), whereas semantic and syntactic speech errors did not differ based on idiom decomposability, suggesting similar representation and processing (Cutting & Bock, 1997). Thus, greater investigation of the impact of decomposability on processing of idioms is necessary.

1.2 | Electrophysiological correlates of idioms

In order to better understand mechanisms of language comprehension, researchers have recorded brain activity while participants read language stimuli and identified components of event-related potentials (ERPs) that relate to language processing. Here, two main ERP components are of particular interest: the N400, a central negativity between 200 and 600 ms that is elicited by all meaningful stimuli and reflects retrieval from semantic memory and semantic processing (Kutas & Federmeier, 2011), and the P600, a posterior positivity observed between 500–900 ms that may reflect integrative processing, particularly following unexpected sentence information (Brouwer et al., 2017; Kim & Osterhout, 2005). To better understand figurate comprehension, researchers can examine how the amplitudes of these ERP components differ between idiomatic and literal comprehension, reflecting differences in engagement of particular processes.

ERP studies have examined compositional processing in idiom and literal phrases (Canal et al., 2017; Laurent et al., 2006; Liu et al., 2010; Moreno et al., 2002; Rommers et al., 2013; Vespignani et al., 2010). For instance, Rommers et al. (2013) had participants read either idiomatic or literal sentence contexts that had expected, unexpected, or semantically associated target words. ERPs to target words in literal contexts showed a classic graded N400 effect—decreasing amplitudes with increasing expectancy (Federmeier

et al., 2007). For target words in idiomatic contexts, elicited N400 amplitudes were reduced compared to targets in literal contexts; however, the graded N400 effect for expectancy was not observed, and greater P600s were observed for the unexpected targets relative to the expected completion. The authors argued that the lack of an N400 effect for idioms reflected compositional processing “shutting off” during idiom comprehension; namely, retrieval of semantic information from memory did not differ based on expectancy of the target word, suggesting the semantics of the idiom was retrieved as a whole, in line with noncompositional theories of processing. While the P600 effect for idioms was not discussed, it may have reflected a detection of the violation of the idiom, leading to downstream re-analysis. Canal et al. (2017) reported similar results, in that N400 amplitudes did not differ between idiomatic and literal contexts, but a frontal positivity in a similar time window (between 400–600 ms) at the phrase-final word was found in idiomatic relative to literal contexts, an effect that the authors related to semantic reanalysis (Kaan & Swaab, 2003; Van Petten & Luka, 2012). However, the predictability of the stimuli, as well as the additional contextual information afforded by the preceding words of the sentence, may have contributed to these results. Readers can use contextual information to generate predictions about upcoming words, and this leads to reduced amplitude N400s (Federmeier et al., 2007). Additionally, unpredictable information, such as words that violate the built-up event structure, can elicit P600 responses (Kuperberg et al., 2003). In the abovementioned studies, critical words were preceded by rich contextual information, and the correct phrase-final word was predictable given the preceding sentence context in both the literal and idiom conditions. In the idiom condition, once the comprehender recognized the first constituent or two as matching their prediction, they might conduct less rigorous compositional processing, as the figurative meaning can be retrieved from memory, leading to the absence of an N400 expectancy effect. In fact, the figurative meaning may have already been somewhat generated by the preceding sentence context, rendering further processing of the idiom unnecessary. In the literal condition, however, even when expectation was met by the first couple of words, the processor would still continue full compositional processing, as the same notion can be communicated in many literal ways. This hypothesis would suggest that idiomaticity and predictability may interact in their impact on comprehension and amplitudes of ERPs, and may depend on the amount of prior context present, yet this specific interaction has not been tested in previous studies. While previous studies have attempted to match the characteristics such as cloze probability of stimuli across conditions of interest, and have reported inferential statistics demonstrating that the stimuli “did not differ” between conditions, such tests are susceptible to Type II errors

and do not capture the potential influence these variables may still have on processing (Sassenhagen & Alday, 2016).

In addition to ERP analyses, it is informative to investigate how modulations of endogenous neural oscillations at different frequencies relate to cognitive processes through the use of time-frequency analyses (Cohen, 2014). Several studies reported that gamma oscillations (50–100 Hz) play a key role in language comprehension, associating gamma with semantic unification (Hagoort et al., 2004) or pre-activation and processing of predicted words (Lewis & Bastiaansen, 2015; Wang et al., 2018). Two idiom studies examining oscillatory activity reported gamma-band activity changes. Rommers et al. (2013) reported increased gamma (60–70 Hz) at the phrase-final words for literal phrases that ended expectedly relative to idioms that ended expectedly, as well as for literal phrases that ended expectedly relative to unexpectedly. The authors found no differences across conditions containing idioms, and argued that compositional processing was conducted only for literals and not idioms. Similarly, Canal et al. (2017) reported increased gamma (55–70 Hz) power at the second word position in three-word literal phrases embedded in expected contexts relative to idioms, suggesting that compositional processing was not continued once the linguistic input was recognized as idiomatic. Both Rommers et al. (2013) and Canal et al. (2017) associated gamma with more compositional processing in literal than in idiom conditions. However, both of those studies used highly predictive contexts, which means that prediction could be a confound to idiomaticity, given that gamma has been associated with both prediction and idiomaticity. To better understand how gamma band activity may relate to idiom comprehension and predictability, idiomatic and literal phrases with a range of predictability should be presented.

1.3 | The present study

The present study investigated the compositional processing of idioms and literal sentences, and whether predictability and decomposability play a role during processing. Familiar idioms were selected from a normed idiom dataset (Bulkes & Tanner, 2017) and

embedded in sentential contexts with minimal preceding contextual information, e.g., “The group *broke the ice*...” (Table 1). The selected idioms were 3-word idioms with the recognition point (Cacciari & Tabossi, 1988) happening at the 3rd, critical word. The selected idioms also varied in their decomposability ratings, as well as the cloze probabilities of the 3rd critical word. Finally, we also included a condition in which an adjective was inserted right before the critical word, e.g., “The group *broke the ‘freezing’ ice*”. This condition served two purposes. First, the adjective was an unpredicted word at the same position as the predicted target word, and thus responses to expectancy violations in idiomatic and literal contexts could be compared, as in Rommers et al. (2013). Second, if the configuration of an idiomatic expression—regardless of its decomposability—is key to its recognition and holistic processing, then any disruption to that configuration should result in undertaking the default, fully compositional processing route. Thus, when examining the target word, idioms with adjective insertions may elicit ERP patterns similar to literal strings. If, however, this notion of the configuration is more flexible and the general “gist” of an expression is all that is needed to trigger figurative meaning activation, then idioms containing adjective insertions will not differ from idioms without insertions. Given that standard compositional processing was expected for literal strings, we expected no difference in ERP amplitudes between literal strings with and without adjective insertions.

We analyzed ERP amplitudes at the target words as well as the pre-critical words (the word prior to the critical word). Based on the literature (Canal et al., 2017), we hypothesized that idioms will have less compositional processing than literal sentences, which means that critical words in idiomatic phrases would show a reduced N400 amplitudes compared to literal contexts. We also expected predictability to influence the processing of idiomatic and literal phrases, leading to an interaction of these factors. Given the mixed results in the literature, we examined both the N400 and the P600 time windows. We also expected that adjective insertions would disrupt processing of idioms and induce greater compositional processing, influencing N400 amplitudes elicited by critical words, but literal phrases

Condition	Example
Idiom	The group broke the <u>ice</u> and introduced themselves
Idiom-Adjective	The group broke the <i>freezing</i> <u>ice</u> and introduced themselves
Literal	The group slipped on the <u>ice</u> and all fell down
Literal-Adjective	The group slipped on the <i>freezing</i> <u>ice</u> and all fell down

Note: Critical word is underlined. Precritical inserted adjective is in italics.

TABLE 1 Example stimuli.

would be largely unaffected. Finally, we expected that N400 responses to adjective insertions would be larger than critical words in idiomatic contexts, unlike what was found by Rommers et al. (2013), as the minimal preceding sentence context in our stimuli would not lead to semantic analysis shutting off. Similarly, for the time-frequency analysis results, we expected to observe increased gamma-band power for target words in literal contexts compared to idiomatic contexts, as Rommers et al. (2013) and Canal et al. (2017) reported, but that this effect would be influenced by the predictability of the critical word, as gamma-band activity may reflect predictive processing.

Our design was novel in that we presented idioms that varied in decomposability, as no study to date has examined how idiom decomposability influences electrophysiological responses during language processing. Given the novelty of this design, we examined how decomposability affected both N400 and P600 amplitudes, as well as gamma-band activity, in a data-driven fashion. Idioms that are more decomposable, or transparent in their form-to-meaning mapping, may elicit engagement of compositional processing similar to literal phrases, whereas non-decomposable or opaque idioms may require more direct memory retrieval to understand the figurative meaning. This difference may manifest as differences in N400 amplitudes, with smaller N400s elicited by decomposable idioms, or as a difference in P600 amplitudes, with larger P600s for non-decomposable idioms. Decomposable idioms may also elicit greater gamma band power, as compositional processing may lead to similar semantic integration processes as in literal language processing.

2 | METHOD

2.1 | Participants

24 right-handed monolingual speakers of American English ($M = 18.6$ years, range: 18–20 years, 10 males) participated in the EEG experiment for course credit. Participants reported no history of seizure or neurological trauma or developmental or reading disorder. All had normal or corrected-to-normal vision. Data from four participants were excluded due to excessive artifact in the raw EEG (over 50% of trials). Data from 20 participants are included in further analyses.

2.2 | Materials and design

A total of 160 idioms were selected from Bulkes and Tanner (2017) with varied decomposable ratings (range

of 0.13–0.90, average = 0.5) and with a mean familiarity rating of 3.5 (range of 2.7–4.5, 5 = very familiar). Stimulus properties of the selected idioms are shown in Figure S1. 160 sentence frames were then created (see Table 1), which could include one of the 160 idioms, a matched literal phrase, an idiom with an adjective inserted, or a matched literal phrase with an adjective inserted. Idiomatic sentences featured a familiar idiom either in its typical configuration (idiom condition) or with an adjectival insertion that was related in meaning to the idiom-final word (idiom-adjective) at the point of the idiom-final word. Literal sentences were either syntactically similar to the Idiomatic sentences (literal condition) or included the adjectival insertion from the idiom-adjective condition (literal-adjective condition) at the point of the phrase-final word. 4 lists were created, made up of 40 sentence stimuli from each of the 4 conditions, that were counterbalanced across participants, such that each of the 4 types of stimuli (e.g., “break the ice”/“break the freezing ice”/“slip on the ice”/“slip on the freezing ice”) were included in the experiment. Sentential contexts up to the phrase-final, or henceforth the critical word, were on average very brief and provided little contextual information, designed not to bias toward the idiomatic or literal interpretation at the point of the critical word. The verbs in the idiomatic and literal conditions were matched for concreteness (3.3 vs. 3.3) using the Ghent concreteness norm. The numbers of words in the contexts prior to the verbs in the idiomatic and literal conditions were similar (4.2 vs. 4.5); these numerical differences were not statistically significant. The list of stimuli can be found in the Supplementary Materials.

Predictability of the stimuli was determined with a separate cloze test. Eighty participants were recruited from the psychology subject pool. The 160 stimulus sentences and their four versions were Latin-square rotated, which resulted in four lists with each list containing only one version of each item. Items were randomized within each list. Each participant was presented with one list, and was instructed to provide the first continuation that came to mind for each of the sentence preambles (e.g., *The group slipped on the _____*). Paired t -tests showed no significant difference in cloze probability between idiom ($M = 0.29$, $SD = 0.30$) and literal ($M = 0.30$, $SD = 0.29$) conditions; $t(159) = .32$, $p = .75$. There was also no significant difference between idiom-adjective ($M = 0.28$, $SD = 0.27$) and literal-adjective ($M = 0.31$, $SD = 0.29$) conditions; $t(159) = .89$, $p = .37$. Additionally, there was no significant correlation between cloze probability and idiom decomposability in both the idiom ($r = .004$) and idiom-adjective condition ($r = .1$). However, given that these were variables of theoretical interest and inferential statistics on stimulus characteristics may be inappropriate for truly

determining differences (Sassenhagen & Alday, 2016), we analyzed neural responses across the range of cloze probability and decomposability.

2.3 | Procedure

Participants first gave informed consent and completed a language background questionnaire. They were then fitted with an EEG electrode cap (30 min) and seated 80 cm away from a LCD screen in a sound attenuated booth. To begin, an experimenter read the instructions to the participant, after which the participant completed a short practice. In the experiment, each trial began with a 300 ms fixation cross in the center of the screen. Sentences were presented word-by-word in a rapid serial visual presentation. Non-critical context words were presented for 250 ms each with a 300 ms blank screen between each word. Critical words were presented for a period of time determined by critical word length up to 400 ms ($M = 292$ ms, range = 211–400 ms). The varied time for critical words was used to reduce the influence of word offsets and word length on ERPs (Coulson & Van Petten, 2002; Hauk & Pulvermüller, 2004). Participants were asked to read each sentence normally for comprehension. After each sentence, they were asked to judge as quickly and accurately as possible whether the sentence they just read was acceptable by pressing a “good” or “bad” button on a response box. Response button (left/right) was counterbalanced across participants. The acceptability judgment was followed by a “Blink, and continue” screen, where participants could blink, stretch, or move around before proceeding to the next trial. Sentences were presented in four blocks, where participants had the option of taking a short break between blocks to minimize fatigue. Scalp impedances were checked in between blocks. The entire task took approximately 35–45 minutes.

2.4 | ERP data processing & analysis

Continuous EEG was recorded from 64 Ag/Ag-Cl electrodes mounted in an elastic cap in accordance with the extended 10–20 system (Jasper, 1958) using an actiCHamp bioamplifier system (Brain Products, Germany). Data were referenced to a vertex electrode (Cz) during recording. Impedances were kept below 10 k Ω . EEG was amplified and digitized at 500 Hz with a 100 Hz high cut-off filter and a 10 s time constant (0.016–100 Hz band-pass).

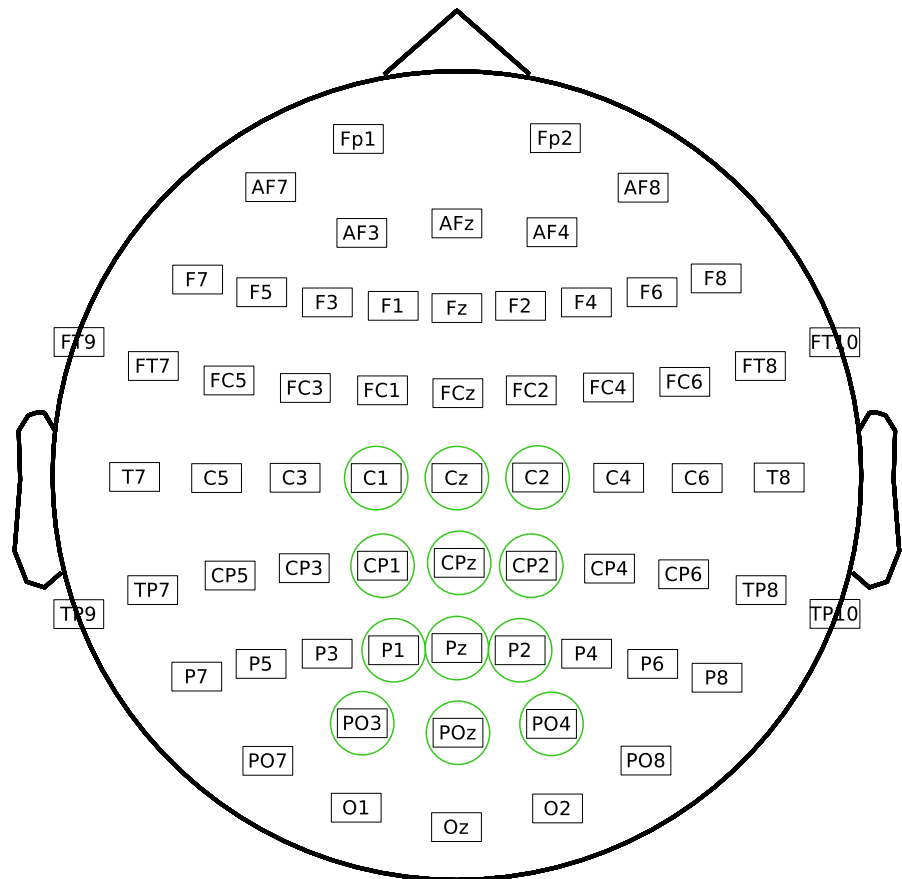
Offline, data were processed using EEGLAB (Delorme & Makeig, 2004) and ERPLAB (Lopez-Calderon &

Luck, 2014) toolboxes in Matlab. Data were filtered with a 4th order 0.1 to 30 Hz bandpass Butterworth filter (–6-dB cutoff, 12-dB/octave roll-off), down-sampled to 200 Hz, and re-referenced to the average of the left and right mastoids. Eyeblinks were removed using AMICA (Delorme et al., 2012). Data containing extreme excursions exceeding ± 80 microvolts were rejected. Data were subsequently visually inspected for blinks, saccades, drift, skin potentials or other artifacts, and individual trials that were contaminated by these artifacts were rejected. Data with 50% or more trials rejected from any one experimental condition were excluded ($N = 4$). In the remaining data ($N = 20$), an average of 5.66% of trials were excluded. ERPs were averaged relative to a 200 ms pre-stimulus baseline, time-locked to the onset of the critical word and pre-critical word (underlined in Table 1). The 64-channel montage, as well as the cluster of channels selected for plotting ERPs, is presented in Figure 1.

Our interest in this study was the influence of predictability and decomposability on neural processing of idiomatic and literal phrases. Therefore, we used a linear mixed-effect modeling approach of single trial measurements to analyze the ERP differences while taking into account the variables of interest (Bates et al., 2015). A central-posterior region of interest was created by averaging together 16 channels (shown in Figure 1). For each trial, average amplitudes were extracted from this ROI in two time windows: 300–500 ms (to examine the N400) and 600–800 ms (to examine the LPC). These signals were then entered into linear mixed-effect models as dependent variables, with fixed effects of idiomaticity (literal or idiom), adjective insertion (inserted or not), cloze probability, and decomposability. Categorical fixed effects were contrast coded, i.e., set to –0.5 and 0.5 (Brehm & Alday, 2022). Random intercepts were included for participants and items, as well as random slopes for participants for fixed effects of interest. Significance testing of fixed effects were estimated with t -tests using the Satterthwaite method in the lmerTest package (Kuznetsova et al., 2017).

Planned comparisons were as follows: We compared idiom to literal conditions at the critical word (e.g., *ice*) to examine the effect of idiomaticity alone, and included cloze probability as a predictor to examine the interaction of predictability and idiomaticity. We also compared idiom-adjective to literal-adjective conditions to examine the effect of the added, unexpected word in both literal and non-literal environments both at the point of the insertion (the pre-critical word, *freezing*) and the phrase-final critical word (*ice*). We further included decomposability as a predictor to investigate the effect of decomposability. We conducted separate analyses to predict the amplitude of the N400 and the P600.

FIGURE 1 Electrode layout and central ROI used for single-trial analysis and plotting ERPs.



2.5 | Time frequency data processing & analysis

For time-frequency analyses, the EEG data were reprocessed. First, the raw data was filtered with a 2nd order 1 to 100 Hz bandpass Butterworth filter. After re-referencing to the average mastoids, the data were split into epochs with 500 ms pre-stimulus and 1200 ms post-stimulus. As with the ERP analysis, eyeblink correction was applied to trials containing blink artifacts with AMICA-based component rejection. A final moving-window artifact scan identified and removed trials containing amplitudes exceeding ± 80 microvolts.

Time-frequency decomposition was performed using FieldTrip functions (Oostenveld et al., 2011). EEG epochs were convolved with Morlet wavelets starting with a width of 5 and increasing linearly to a width of 11 across a frequency range of 20 to 100 Hz, resulting in time-frequency bins of 20 ms and 2 Hz, respectively. A baseline period of -400 to -100 ms was used for relative baseline correction following data averaging. All statistical analyses of time-frequency data were carried out with non-parametric cluster-based permutation tests (Maris & Oostenveld, 2007).

We intended to examine the effects of cloze probability and decomposability on oscillatory activity in a

continuous fashion, similar to the analysis of ERPs—however, choosing a time-window, set of channels, and frequency range a priori was more difficult. Therefore, we used correlation cluster-based permutation tests to perform the analysis with a more data-driven approach. Here, time-frequency-channel data were averaged across subjects for each item, and the averages were baseline corrected. To examine the effect of predictability, a correlation between the power and the cloze probability ratings for each item was computed at each time-frequency-channel bin, resulting in correlation coefficients (r -values) at each bin, and clusters were created by grouping adjacent bins that had resultant p -values $< .05$. The r -values of observed clusters were summed to create cluster-level statistics, and these statistics were compared to a permutation distribution that was created by permuting the subject labels and repeating the clustering procedure 1000 times, taking the max cluster statistics each time. Any p -values $< .05$ represent a statistically significant relationship between power and cloze probability in the observed time-frequency-channel cluster. This analysis was also run using decomposability values to examine the effect of decomposability.

Additional analyses were performed for each of the previous analyses to discriminate between phase-locked

and non-phase-locked time-frequency responses (Kalcher & Pfurtscheller, 1995), as phase-locked responses are often captured in the recorded ERP. Namely, for each item, the average ERP across subjects was subtracted from each subject prior to the time-frequency analysis, and the correlation analyses were performed on the ERP-subtracted data. Results from these analyses are not reported here, as they ultimately did not differ from the reported effects and did not change the drawn interpretations.

3 | RESULTS

3.1 | Behavioral results

A repeated-measures ANOVA was conducted on participants' acceptability judgments, with within-participant factors of idiomaticity (Idiom vs. Literal) and adjective-insertion (Adjective vs. No Adjective). This revealed a significant interaction between idiomaticity and adjective-insertion ($F(1,19) = 6.37, p = .02$), as well as main effects of idiomaticity ($F(1,19) = 25.27, p < .01$) and adjective-insertion ($F(1,19) = 21.66, p < .01$). The mean acceptability rating was the highest for the Literal condition (77% Good), and was reduced for the other conditions (Idiom, 71%; Literal-Adjective, 71%; Idiom-Adjective, 51%). Literal sentences had higher acceptability than Idiom sentences, and adjective insertions led to reductions in acceptability; importantly, adjective insertion led to a greater reduction in acceptability for Idioms than for Literal sentences.

We conducted an additional analysis to examine the influences of predictability and decomposability on acceptability ratings for idiomatic phrases. A mixed logistic regression model was used to predict trial-level acceptability ratings (1 = acceptable, 0 = unacceptable), including fixed effects of adjective-insertion, predictability, and decomposability, with random intercepts and slopes for participants (Jaeger, 2008). Significant effects were found for adjective-insertion (Wald's $z = 3.73, p < .01$), confirming the results of the ANOVA, as well as decomposability (Wald's $z = 3.00, p < .01$), but there was no significant effect of predictability (Wald's $z = 0.36, p = .72$), and no significant interactions. Thus, adjective insertions influenced acceptability ratings, and more decomposable idioms were rated as more acceptable.

3.2 | ERP results

Grand averaged ERP waveforms at pre-critical and critical words are in Figure 2.

At the pre-critical word, analyses were focused on the N400 time window, as the critical word overlapped with the P600 time window. When fixed effects of idiomaticity and adjective insertion were included in the model, a main effect was found for idiomaticity ($t = -2.07, p = .047$), as well as for adjective insertion ($t = 2.90, p = .01$), but there was no significant interaction ($t = 1.38, p = .17$), suggesting that pre-critical words in idiomatic phrases elicited more negative N400 amplitudes, and adjective insertions elicited more negative N400 amplitudes. While the interaction between idiomaticity and adjective insertion was not significant, the experimental design warranted examining the simple effects, as effects were only expected to emerge when adjectives were inserted. When focusing on the phrases without adjective insertions, no significant effects were found at the pre-critical word. In contrast, when focusing on the phrases with adjective insertions, a main effect of idiomaticity was found ($t = -2.27, p = .02$); namely, the Idiom-Adjective condition elicited more negative amplitudes than the Literal-Adjective condition. The effect of cloze probability (of the upcoming critical word) was also significant ($t = -2.92, p < .01$), demonstrating that when the critical word was more predictable, the adjective insertion elicited a more negative N400. The interaction of idiomaticity and cloze probability was not significant ($t = 1.01, p = .31$). Additionally, idiom decomposability was not a significant predictor of N400 amplitude ($t = 0.75, p = .46$). Thus, adjective insertions led to more negative N400s, particularly when the critical word was predictable, and adjective insertions into idiomatic phrases elicited more negative N400s than for literal phrases.

At the critical word, separate analyses were conducted on phrases with and without adjective insertions, as the critical words for phrases with adjective insertions were at a later sentence position, leading to a more positive shift (see Figure 2a). For phrases without adjective insertions, there was a significant effect of cloze probability ($t = 2.59, p = .02$) in the N400 time window, with more predictable words eliciting more positive N400 amplitudes, but no significant effect of idiomaticity ($t = 0.62, p = .54$) or idiom decomposability ($t = -0.37, p = .71$). In contrast, for phrases with adjective insertions, significant effects of idiomaticity ($t = 2.97, p < .01$) and cloze probability ($t = 2.08, p = .04$) were found in the N400 window, though there was no significant interaction ($t = -1.66, p = .10$) or effect of decomposability ($t = 0.87, p = .39$). Thus, cloze probability predicted N400 amplitudes regardless of idiomaticity, whereas idiomaticity predicted N400 amplitudes only for phrases with adjective insertions.

Turning to the P600 time window, a striking difference was found compared to the N400 window. Namely, for phrases without adjective insertions, there was no

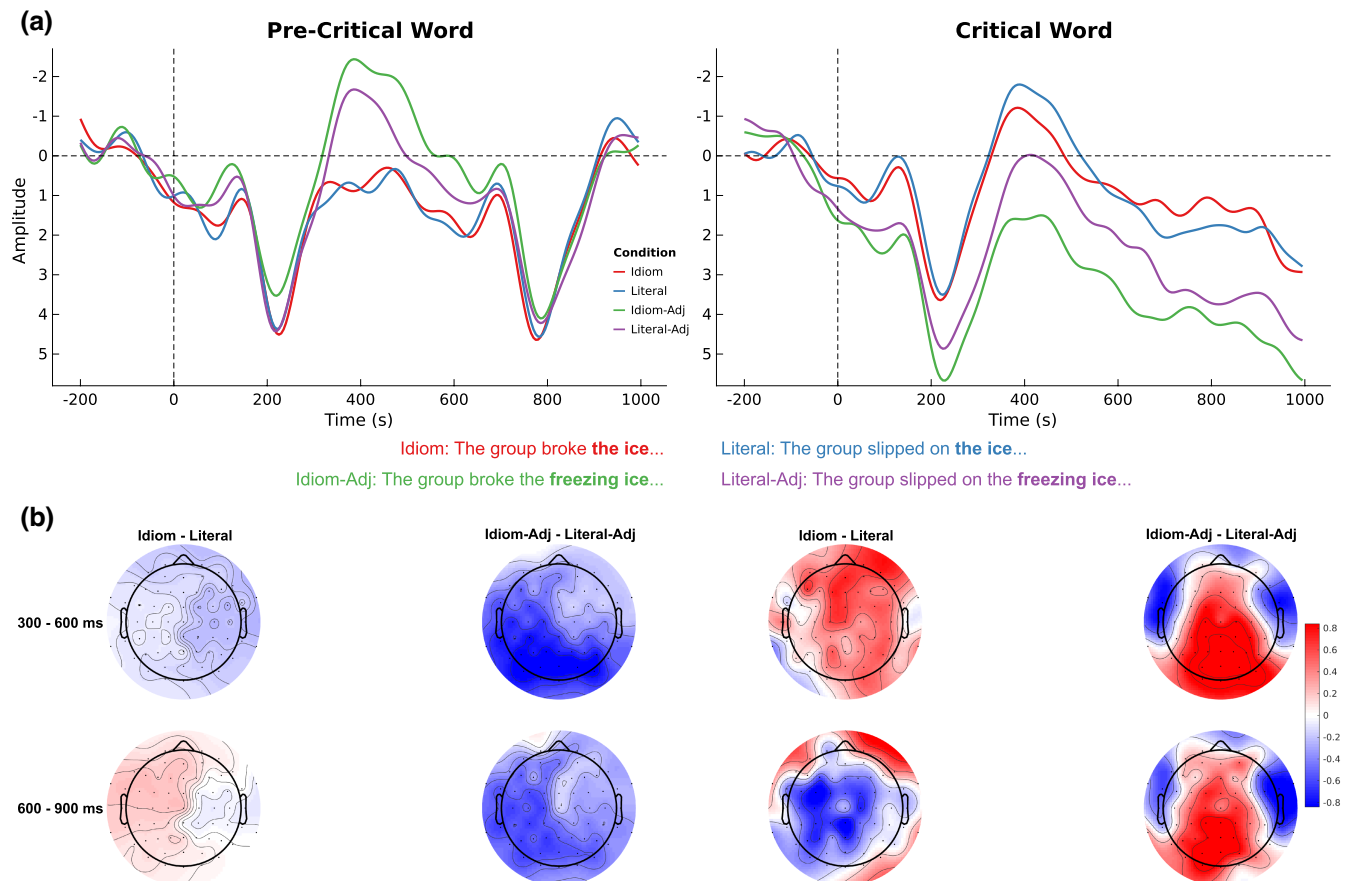


FIGURE 2 (a) ERPs at the pre-critical word (left) and the critical word (right) pooled from 12 central-parietal electrodes. The 4 conditions are Idiom (red), Literal (blue), Idiom-Adjective (green), Literal-Adjective (purple). (b) Topography plots of mean amplitude differences between conditions (Idiom—Literal, Idiom-Adjective—Literal-Adjective) in the 300–600 ms time window (top row) and in the 600–900 ms window (bottom row). ERPs were filtered with 10 Hz low-pass filter for visualization, not for statistical comparisons.

significant effect of cloze probability ($t = -0.50$, $p = .62$) on the P600, yet there was a significant effect of idiomaticity ($t = -2.29$, $p = .02$) as well as a significant interaction between idiomaticity and cloze probability ($t = 2.02$, $p = .04$). Here, cloze probability did not predict P600 amplitudes to critical words in literal phrases, but did predict P600 amplitudes to critical words in idiom phrases, with more predictable words eliciting more positive P600s. This effect can be seen in Figure 3. Idiom decomposability did not predict P600 amplitudes ($t = 0.52$, $p = .60$). Thus, predictability modulated the amplitude of the P600 component, but only for idiomatic phrases. For phrases with adjective insertions, significant effects were not found in the P600 time window at the critical word, potentially suggesting the adjective insertion altered the processing that would normally occur in an idiomatic phrase.

The next analysis focused on the effect of unexpected adjective insertions on ERP amplitudes for both literal and idiomatic phrases. We compared critical words from phrases without adjective insertions to pre-critical adjectives in the adjective insertion conditions—the same word position in the sentence—for both idioms and literal

phrases. Here, the adjective insertion essentially served as an unexpected word in the same position as the expected word. Grand average ERPs to critical and pre-critical words are displayed in Figure 4. In the adjective-insertion conditions, the onset of the final word overlapped with the P600 time window; therefore, analyses were focused on the N400 time window. A linear mixed-effect regression model with fixed-effects of idiomaticity (Idiom vs. Literal) and adjective (Adjective vs. No Adjective) found no significant effect of adjective ($t = -0.75$, $p = .45$), but found a significant effect of idiomaticity ($t = -2.28$, $p = .02$) and a significant interaction ($t = 2.56$, $p = .01$). Thus, N400 amplitudes differed between critical words and adjective insertions for idioms, but not for literal phrases.

3.3 | Time-frequency results

3.3.1 | Cloze probability: Correlation analysis

Only statistically significant clusters are reported. We first examined if predictability influenced gamma-band

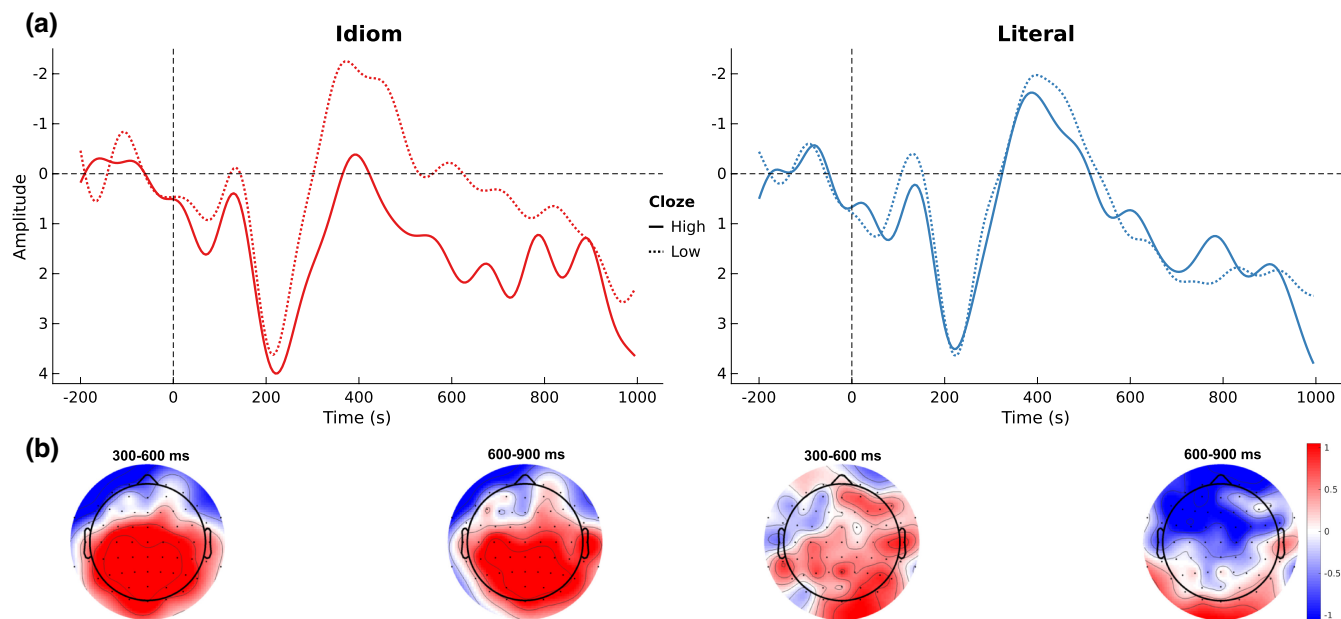


FIGURE 3 (a) ERPs at the critical word pooled from 12 central-parietal electrodes, separated into a high cloze bin (solid lines) and a low cloze bin (dotted lines). The conditions are Idiom (left) and Literal (right). (b) Topography plots of mean amplitude differences between cloze bins (High—Low), in two time windows and for both conditions.

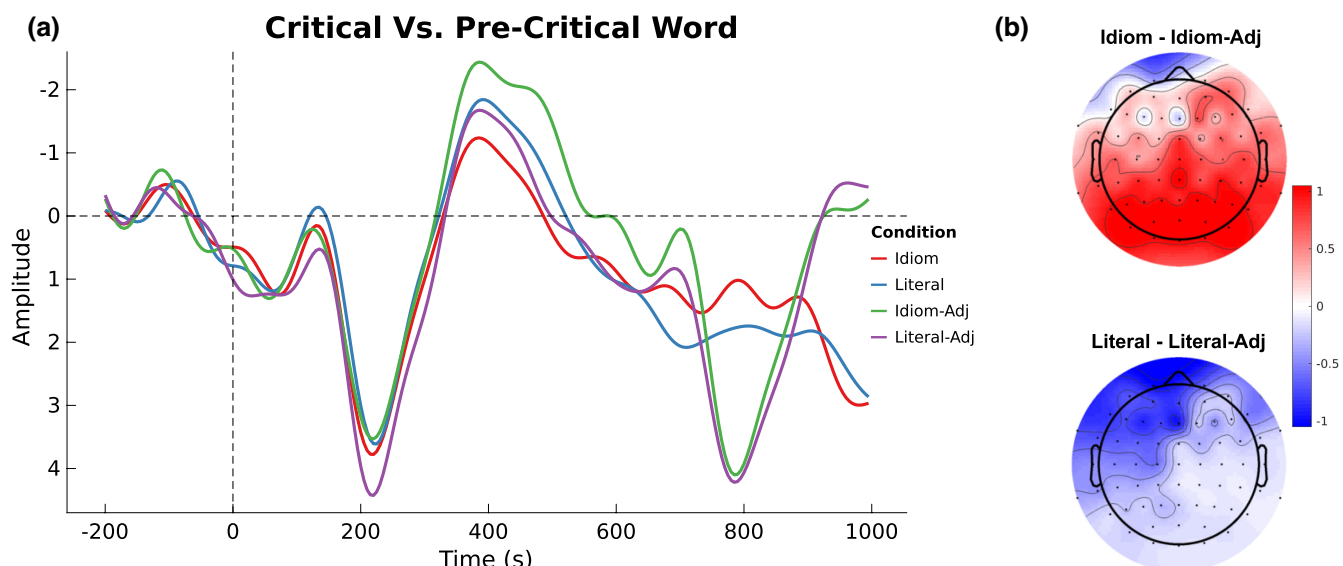


FIGURE 4 (a) ERPs at the critical word for non-adjective conditions, and to the pre-critical word for adjective insertion conditions, pooled from 12 central-parietal electrodes. The 4 conditions are Idiom (red), Literal (blue), Idiom-Adjective (green), Literal-Adjective (purple). (b) Topography plots of mean amplitude differences between conditions (Idiom—Idiom-Adjective, and Literal—Literal-Adjective), in the N400 time window.

activity, as previous studies have shown. To do so, an item-level analysis was conducted in which neural activity was correlated with cloze probability of critical words, and multiple comparisons were corrected with permutation testing. A significant cluster was found; namely, gamma power was significantly positively related to cloze probability for critical words in idiomatic phrases ($p = .01$), i.e., gamma power increases as cloze probability (Figure 5). The observed cluster time range was

from roughly 400 to 700 ms post-stimulus, and ranged in frequency from 60–100 Hz. The cluster had a central-posterior topography, with a slight left hemisphere bias. Average oscillatory power in this cluster was correlated with cloze probability with $r = .6$. Interestingly, we found no significant relationship between oscillatory power and cloze probability for critical words in literal contexts, or for critical words in idiomatic contexts with adjective insertions.

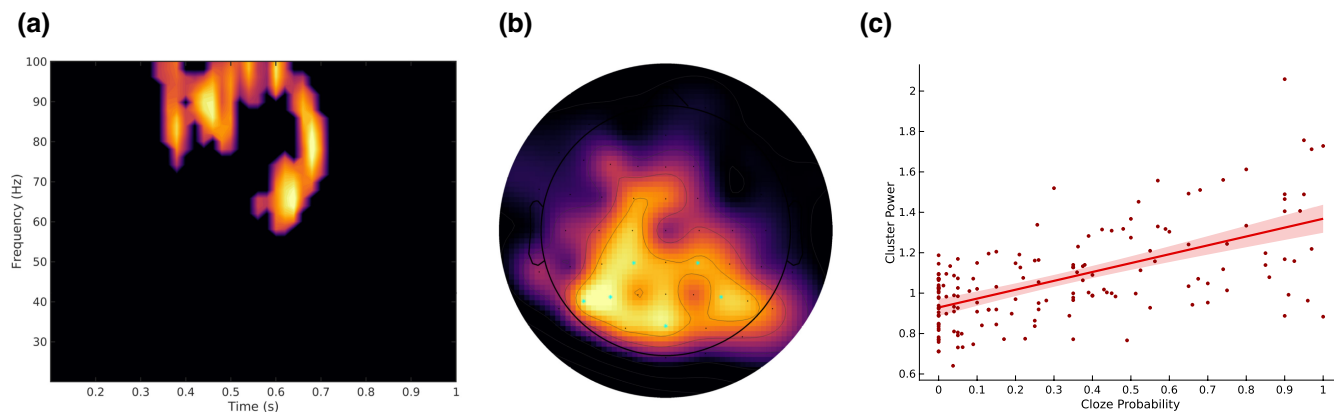


FIGURE 5 Results for correlation cluster analysis with idiom time-frequency activity and cloze probability. (a) Time-frequency plot depicting cluster of significant time-frequency bins, with time on the x axis, frequency on the y axis, and the color representing t -value. (b) Topography of the significant cluster. Highlighted channels show the maximum of the cluster, which were averaged for creating plot A. (c) Correlation between cloze probability and oscillatory power derived from the significant cluster for each idiom, averaged across participants ($r = .60$).

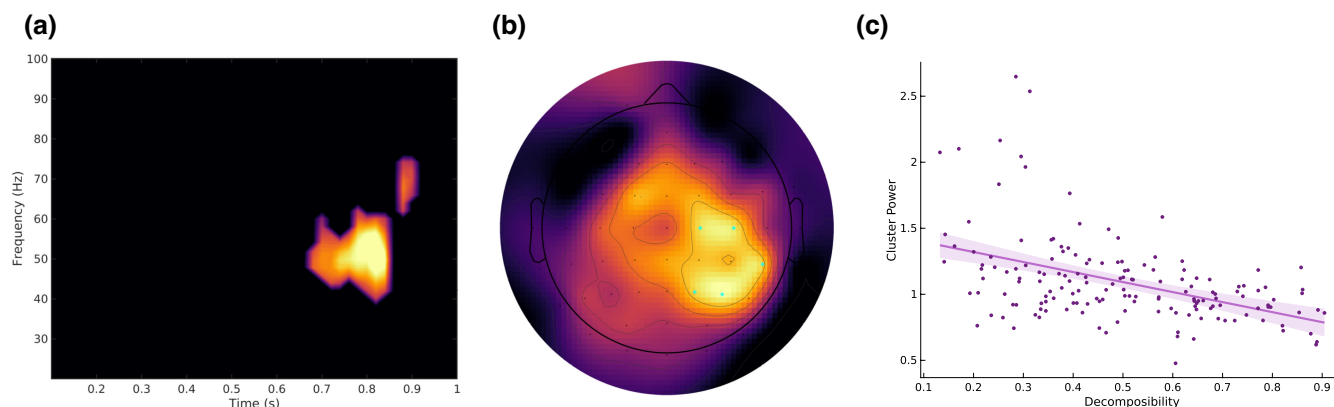


FIGURE 6 Results for correlation cluster analysis with idiom time-frequency activity and decomposability. (a) Time-frequency plot depicting cluster of significant time-frequency bins, with time on the x axis, frequency on the y axis, and the color representing t -value. (b) Topography of the significant cluster. Highlighted channels show the maximum of the cluster, which were averaged for creating plot A. (c) Correlation between decomposability and power derived from the significant cluster for each idiom, averaged across participants ($r = -.46$).

3.3.2 | Decomposability: Correlation analysis

We next ran a similar analysis, but correlating oscillatory power with idiom decomposability. A significant cluster identified that gamma power was significantly negatively related to decomposability for idioms ($p = .01$), i.e., gamma power decreased as decomposability increased (Figure 6). The observed cluster time range was from 0.66 to 0.9 s. The gamma effect was also somewhat lower in frequency than the previously described correlation cluster with cloze probability, ranging from 36 to 74 Hz, and had a different topography, with a right hemisphere bias. There was no significant relationship between power and decomposability in the idiom-adjective condition, suggesting that the idiom's particular configuration had an impact on the degree to which gamma band activity was elicited.

4 | DISCUSSION

We examined compositional processing during the comprehension of idioms with and without an adjective insertion, and how predictability and decomposability influence this process. Participants read sentences containing idioms (*The group broke the ice ...*), their literal match (*The group slipped on the ice ...*), and those same sentences with adjective insertions (*The group broke the freezing ice ...*; *The group slipped on the freezing ice ...*). The items varied in predictability of the critical words and decomposability of the idiom, allowing us to examine how these factors separately influenced processing of figurative and literal language. We collected EEG as participants read sentences in order to analyze differences in amplitudes of ERP components, as well as gamma-band oscillatory

power, between target words in idiomatic and literal language contexts.

At the phrase-final position—the critical word, words in idiomatic contexts showed a reduced N400 compared to words in literal contexts, and this N400 difference between idiomatic and literal contexts became larger when an adjective was inserted in the phrase. Importantly, the cloze probability of the critical word predicted N400 amplitudes in both literal and idiomatic phrases. This result suggests that readers still engaged in compositional processing of idiomatic phrases and did not immediately process them in a non-compositional manner; namely, if the figurative meaning of the idiom was retrieved from memory and compositional processing was not engaged, then the predictability of the idiom's critical word would have little effect on the N400. Though, it remains possible that the kinds of compositional processing engaged by idiomatic versus literal contexts differed, given that N400 amplitudes were overall reduced for critical words in idiomatic phrases compared to literal phrases. Additionally, if processing of idioms was solely non-compositional, then disrupting the idiom with an adjective would have little effect, but we observed that disrupting the phrase with an adjective led to facilitation of semantic memory access of the target word. This result differs from Canal et al. (2017), where they reported no N400 difference between idiomatic and literal contexts, but instead a frontal positivity (400–600 ms). One potential explanation is that the effect we observed is the same as their frontal effect, as the time windows overlap, although the topographies differ to some degree. Alternatively, this discrepancy may be due to the difference in stimulus materials used between studies. Both Canal et al. (2017) and Rommers et al. (2013) presented sentences that provided predictive contextual information prior to the critical word, and their critical words had very high cloze probability. When readers have a rich context to draw meaning from prior to the appearance of an idiom, then they may engage non-compositional processing to a greater degree. Here, our context was on average not highly predictive (e.g., “He got his hands *dirty* and got to work”). In other words, due to our attempt to minimize contextual information prior to the critical phrase, participants may not have accessed the figurative meaning of the idiomatic phrase to the same degree as in other work where the context biases toward an idiomatic interpretation.

Our analysis of the effect of cloze probability on ERP amplitudes revealed a striking difference between literal and idiomatic processing in a later time window; namely, the P600. A significant relationship between cloze probability and P600 amplitude was only found for critical words in idiomatic contexts, not in literal ones. Thus, only critical words in idiomatic contexts modulated

neural responses outside of the N400 time window such as the P600, a somewhat similar finding to Rommers et al. (2013) and Canal et al. (2017). These results suggest that semantic processing—as indexed by the N400—may be similar between idiomatic and literal language processing, but re-analysis—as indexed by the P600—may be differentially engaged between these two types of language. Additionally, the degree to which compositional analysis of figurative language is engaged may be highly dependent on the amount of preceding contextual information present, as well as the predictability of the constituent words. This and other work (DeLong et al., 2014; Hoeks et al., 2004) highlight the importance of predictability during language comprehension, and that future work specifically manipulating the amount of preceding contextual information prior to a particular word or phrase is critical to understanding the role of context in figurative language processing.

At the pre-critical words, while ERPs did not differ between idiomatic and literal contexts, N400 differences were observed for the adjective-insertion conditions. Specifically, adjectives in idiomatic contexts (... broke the **freezing** ...) elicited larger N400s than adjectives in literal contexts (... slipped on the **freezing** ...). One explanation is that adjectives were more unexpected in an idiomatic context, thus leading to a larger N400 amplitude. However, while cloze probability was a significant predictor of N400 amplitudes, there was also a significant effect of idiomaticity, suggesting predictability was not the sole cause of this effect. On the other hand, this result is also in line with other studies examining words preceding critical words that differ in how predictive they are toward a certain critical word (Freunberger & Roehm, 2017; Maess et al., 2016). Here, when pre-critical words increased the predictability of the following target words, the N400 elicited by the pre-critical word was more negative. This larger N400 amplitude may reflect greater activation of resources for the upcoming target word, leading to greater facilitation at the time of the critical word. Importantly, this is once again in line with a compositional view over a non-compositional view; the adjective insertion did not impede the retrieval of the idiomatic phrase from memory, but instead influence the processing of the final word of the phrase, suggesting word-by-word processing.

We also compared N400 amplitudes elicited by adjective insertions to those elicited by critical words when no adjective was inserted. Here, the adjective insertions serve as unexpected words in the same sentence position as the critical words. We found that adjectives elicited larger N400 amplitudes than critical words in idiomatic contexts, but N400 amplitudes elicited by adjectives did not largely differ than those elicited by critical words in literal contexts. Here, we essentially found the opposite pattern

of the effects reported in Rommers et al. (2013), where cloze probability effects on the N400 were found in literal contexts, but not in idiomatic contexts, which suggests that, in our study, compositional processing was engaged during reading of idioms and was not “shut down”. One interpretation of these differences is that, in Rommers et al. (2013), the figurative meaning of the idiom was to some degree activated by the prior context, and, since predictability was high, compositional processing of the idiom was not engaged, leading to no N400 differences between expected and unexpected words. In contrast, when contextual information is low and the figurative meaning of the idiom cannot be immediately retrieved from memory, compositional processing of idioms may be engaged. An alternative explanation is that the difference in stimuli between studies could contribute to the different effects observed. The stimuli in Rommers et al. (2013) contained unexpected words that could have been semantically related or unrelated to the expected word, whereas the unexpected words in this study were adjectives prior to the critical word. In literal language processing, adjectives related to expected critical nouns may not produce as large N400 violation effects compared to unexpected nouns (Szewczyk et al., 2022; Szewczyk & Wodniecka, 2020). This may not be the same for idiomatic language, where there may be more specific expectations of the class of the upcoming word.

Turning to the time-frequency analyses, we sought to examine how oscillatory activity was related to predictability of critical words. We found a significant relationship between gamma-band activity and the cloze probability of the critical word in idiomatic contexts, with more gamma power for more predictable critical words, in line with previous work (Wang et al., 2012). Additionally, the time frame of the observed cluster was similar to the gamma power difference for expected and unexpected words, as well as literals versus idioms, found by Rommers et al. (2013), although the range of frequencies differed between our studies. Additionally, we did not find the same relationship for critical words in literal contexts. Our results, in conjunction with the ERPs, are more in line with a prediction-based account of gamma oscillations (Lewis & Bastiaansen, 2015); namely, idioms with little surrounding context may show greater prediction effects than literal language, but when a rich context is provided (Rommers et al., 2013), this difference may be reduced. Additionally, their inclusion of unexpected sentence endings might have influenced the degree to which individuals predicted, as well as the type of predictions they generated, leading to differences in gamma band activity. Thus, it would be worthwhile to examine differences in gamma power for figurative processing when individuals are presented only with expected completions versus

when unexpected completions are also included, holding the predictability (cloze probability) constant.

We also examined how decomposability of idioms influenced compositional processing. Decomposability predicted the acceptability ratings of idiomatic phrases, with more decomposable idioms rated as more acceptable. Surprisingly, our regression models found no relationship between ERP amplitudes and idiom decomposability. This suggests compositional processing, at least evidenced by the semantic retrieval indexed by the N400, did not differ based on decomposability of the idiom. It is possible decomposability effects on the N400 could be observed with different task demands or stimuli; for instance, Libben and Titone (2008) found that idiom decomposability only facilitated processing in certain tasks requiring a semantic judgment, and interacted with the familiarity of the idiom. However, we found a negative relationship between gamma band activity and decomposability, with higher gamma band power for lower decomposability idioms. Given the lack of high precision of the results of cluster-based permutation tests (Sassenhagen & Draschkow, 2019), it is not clear if the frequency band of this decomposability effect substantially differed from the previously discussed gamma effect related to predictability. On the other hand, this effect certainly occurred later in time and exhibited a substantially different topography than the gamma band predictability effects, suggesting recruitment of a different process. One possibility is that this specific gamma-band activity may be more related to processing the figurative aspect of language. A second possibility is that the gamma power effects observed in our study may reflect prediction or semantic integration, depending on the timing and topography. In an idiom with low decomposability, the idiom's component parts do not contribute semantically to the figurative meaning of the idiom, and thus semantic integration is more difficult, possibly resulting in greater gamma activity to overcome this difficulty. Alternatively, this gamma band activity could reflect memory retrieval of the idiom's figurative meaning, which is in higher demand when the figurative meaning cannot be obtained from the component parts of the idiom. This explanation is corroborated by our finding that the relationship between gamma-band activity and decomposability was not observed when the idiom was disrupted with an adjective, which may have induced greater compositional processing and less retrieval of the figurative meaning of the idiom. A useful follow-up would be to examine if gamma power in this time is related to magnitude of cross-modal priming, as one study found that idiom decomposability was related to the degree to which the auditorily presented idiom primed responses to visually presented targets related to the figurative meaning, with

decomposable idioms interfering with priming (Titone & Libben, 2014). A relationship between gamma band activity during idiom processing and behavioral priming would suggest some link between this activity and figurative meaning retrieval.

As a set, our results are primarily in line with a hybrid model of figurative language processing. Namely, the ERP and oscillatory results suggest that compositional processing can be engaged when reading idiomatic phrases initially, but the extent to which it is engaged may depend on the predictability of the information as the phrase unfolds, as well as the decomposability of the idiom. Our results are not in line with a non-compositional account of idiomatic processing, where idioms are lexicalized and essentially stored as one unit. By this account, varying the predictability of words in the idiom would have little impact on processing of the word, as compositional processing would not be engaged. Additionally, the inclusion of an adjective in the middle of an idiomatic expression did not impair processing, but instead seemed to increase the observed facilitation compared to a literal expression, suggesting that idioms are not always retrieved from memory as a single unitary chunk of information. Somewhat surprisingly, neither N400 nor P600 amplitudes differed based on idiom decomposability, which may be in line with more strictly compositional models, in that the level of compositional processing did not seem to change. However, different levels of idiom decomposability were associated with gamma band activity, but in a later time window than gamma activity associated with predictability. This dissociation suggests that gamma-band activity at different time-points is associated with separable mechanisms of language processing. These separate mechanisms appear to be engaged differentially based on characteristics of the stimuli and the linguistic context they appear in, which is in line with a hybrid account of processing, in which these features interact to determine how much compositional processing should be done.

AUTHOR CONTRIBUTIONS

Nyssa Bulkes: Conceptualization; data curation; investigation; methodology; writing – original draft. **Vicky Tzuyin Lai:** Conceptualization; funding acquisition; project administration; resources; supervision; writing – review and editing. **Ryan Hubbard:** Formal analysis; Investigation; Methodology; Visualization; Writing – original draft; Writing – review and editing.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1: Supporting Information

Appendix S2: Supporting Information

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